forrestbicker@gmail.com
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Forrest Bicker

Mathematics + Computer Science

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EDUCATION

Harvey Mudd College — Claremont, CA
B.S. Mathematics and Computer Science — *Aug 2021 - May 2025*
· ICPC, Research at AMISTAD Lab, Backgammon Lab

Hunter College High School — New York, NY

EXPERIENCE

Arrowstreet Capital — Boston, MA
Quantitative Researcher — *Jul 2025 - Sep 2026*
· One of 25 researchers on team managing $350B.
· Conducted alpha and risk research.
· Scaled training systems.

DE Shaw & Co. — New York, NY
Quant Intern — *Jun - Aug 2024*
· Researched enhancements to portfolio optimization reward model.
· Used numerical methods to improve quadratic program solver efficiency.

Amazon Web Services — Seattle, WA
Software Engineer Intern — *Jul - Sep 2022*
· Robustified AWS Kubernetes provisioning and recovery systems to orchestrate 10,000-node ultra-scale GPU clusters.

RESEARCH & PROJECTS

Backgammon Research Group — Claremont, CA
Backgammon Research under Dr. Arthur Benjamin — *Nov 2021 - May 2022*
· Designed quantitative strategies for tournament backgammon play under world champion Dr. Arthur Benjamin.
· Developed novel betting strategy that outperforms state-of-the-art methods and proved successful in championship play.

AMISTAD Lab — Claremont, CA
Machine Learning Research under Dr. George Montañez — *May - Sep 2022*
· Proved theoretical bounds on the Bias-Expressivity tradeoff for continuous learning problems and fuzzy membership.
· Published "From Targets to Rewards: Continuous Target Sets in the Algorithmic Search Framework" in ICAART 2024

AWARDS

1st Place Paper at Citadel Global Data Open Championship ($100,000 prize)
1st Place at IMC Prosperity Global Trading Challenge ($25,000 prize, 1st of 13,500)
Top 50 USA at ICPC North American Championship (International Collegiate Programming Contest NAC-NAPC)
1st Place (2x) at West Coast Datathon ($20,000 prize)
1st Place at National Security Agency (NSA) Cybersecurity Hacklytics Challenge
3rd Place Algorithm at Worldwide Correlation One Terminal ($2,500 prize, 3rd of 25,000)

SKILLS

Python, Pytorch, NumPy, Pandas, Statsmodels, CVXPY